

Cherenkov EBM: A Geometric Reasoning Layer for Navigation Between Linear Regimes

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Abstract

We introduce the Cherenkov EBM, a geometric reasoning layer that operates between linear regression contexts. The system does not compete with linear models on linearly separable data; it provides an intermediary navigation layer that connects linear regimes across nonlinear transitions. The energy function is a weighted sum of named geometric primitives drawn from a thirteen-family library inspired by the Chinese Zodiac, with the thirteenth family (linear detachment) modeling directed flow through a topological gap. Primitives are synthesized autonomously from residual structure, with no neural backbone and no fixed catalogue. Augmenting logistic regression with these synthesized features improves held-out accuracy by an average of +16.4 percentage points across eight synthetic geometric benchmarks (5-seed, stratified 70/30 split), with gains up to +46.4pp on concentric circles and +32.4pp on XOR. On real-world tabular datasets where linear regression is already strong, the parsing layer produces near-zero gain, consistent with our central claim that the EBM is a navigation layer between linear regimes rather than a competitor to them. We further introduce V1, a tabular Q-learning protocol that learns which primitive to deploy at each residual state, providing a path toward learned geometric reasoning rather than greedy synthesis.

1. Linear | Nonlinear | Linear

We propose a three-layer computational architecture that distinguishes the Cherenkov EBM from existing parsing layers, kernel methods, and Joint Embedding Predictive Architectures (JEPA):

Layer 1 (linear, stochastic): the raw data, modeled by a standard linear regression. This layer captures the linearly separable structure and produces a residual field.

Layer 2 (nonlinear, geometric): the parsing layer reads the residual field as a geometric signal and synthesizes a sequence of named primitives that explain it. This is the navigation layer.

Layer 3 (linear, augmented): a downstream linear model operates on the augmented feature matrix $[X \mid \phi_1(X) \mid \dots \mid \phi_n(X)]$. The geometry has been made linearly accessible.

JEPA predicts in opaque representation space; Cherenkov predicts in *legible* primitive space. Each primitive has a name, a parameterization, and a geometric meaning. The fingerprint — the ordered list of primitive families retained — is a language-agnostic description of the domain's geometric structure.

The architecture's strength is not in beating linear regression on linear data — that is a category error. The strength is in providing a structured intermediary when the data exhibits piecewise-linear regimes separated by nonlinear transitions. The hypothesis is that real-world geometry is mostly piecewise linear, with a finite alphabet of named transitions stitching the pieces together. The thirteen-family library is our proposed alphabet.

2. The Thirteen-Family Library

Each primitive is a parameterised real-valued function $\phi(x; \theta) \rightarrow \mathbb{R}$. Families are named after the Chinese Zodiac for memorability:

Family	Animal	Geometric meaning
radial	Horse	Isotropic Gaussian bump (kernel-like)
directional	Rooster	Sigmoid along a learned direction
oscillatory	Snake	Sinusoidal along an axis
torsional	Dragon	Angular winding around a center
crossing	Goat	Pairwise feature interaction
trend	Ox	Directed flow with lateral decay
acute	Tiger	Sharp wedge / acute-angle boundary
obtuse	Rabbit	Wide wedge / obtuse-angle boundary
wormhole	Rat	Catenoid throat (topological gap)
saddle	Monkey	Bifurcation saddle (cusp catastrophe)
limit_cycle	Dog	Closed-orbit ring (Hopf bifurcation)
cusp	Pig	Cusp boundary (cusp catastrophe)
linear_detachment	Axolotl	Directed crossing of a topological gap

The Axolotl primitive is novel. It models a directed linear signal passing through a wormhole throat — an approach field $\tanh(\alpha \cdot d_{\text{in}})$, a gap $\exp(-r^2/2\sigma^2) \cdot (1 - \exp(-r^2/2\varepsilon^2))$, and an exit field $\tanh(\alpha \cdot d_{\text{out}})$ where d_{out} is d_{in} rotated by angle θ . The single free angle θ captures how much the coordinate system bends crossing the gap. This is the kind of transition that arises naturally in protein structural class boundaries and neural state transitions.

3. Synthesis Algorithm

Given training data (X, y) , the parsing layer constructs a primitive sequence by greedy residual descent against a target field $T(y) \in \{-1, +1\}$:

1. Initialize residual $R \leftarrow T$.
2. For round $r = 1, 2, \dots, R_{\text{max}}$:
 - 2a. If $\text{Var}(R) < \varepsilon$, terminate.
 - 2b. Sample center candidates from residual-weighted KMeans.
 - 2c. For each family $\in \{13 \text{ zodiac families}\}$, grid-search over parameters and select the (family, params) pair maximizing $|\text{corr}(\phi(X), R)|$.
 - 2d. If the best correlation is below threshold τ , terminate.
 - 2e. Add the primitive. Refit all weights via OLS: $w = \text{argmin}_w \|T - \Phi w\|^2$. Update $R = T - \Phi w$.
3. Return the primitive list (the fingerprint) and weights.

Convergence. Residual variance is non-increasing by construction (OLS guarantees variance reduction or no update). The library is finite, the residual is bounded, so the loop terminates.

4. Validated Results — Synthetic Geometric Benchmarks

Protocol: 5 seeds, stratified 70/30 train/test split, 20-shot subset of the training fold, OneVsRest for multiclass. Baseline is logistic regression. Augmented is the same logistic regression on $[X \mid \Phi(X)]$ where Φ is the fingerprint synthesized on the full training set. Reported is held-out test accuracy.

Domain	LR(X)	LR(X+EBM)	Δ	Top primitives
Circles	51.3% $\pm$ 3.3%	97.7% $\pm$ 3.0%	+46.4pp	radial, limit_cycle
Moons	85.2% $\pm$ 2.3%	97.7% $\pm$ 1.6%	+12.5pp	trend, radial
XOR	50.4% $\pm$ 3.9%	82.8% $\pm$ 8.3%	+32.4pp	crossing, saddle
Saddle	55.9% $\pm$ 2.5%	84.9% $\pm$ 1.4%	+29.1pp	saddle, trend
Cusp	79.6% $\pm$ 4.0%	87.5% $\pm$ 3.6%	+7.9pp	cusp, directional
Iris 3-class	89.3% $\pm$ 5.7%	92.0% $\pm$ 4.1%	+2.7pp	directional, radial
Wine 3-class	96.3% $\pm$ 1.2%	96.7% $\pm$ 0.7%	+0.4pp	directional
Breast Cancer	95.1% $\pm$ 3.0%	95.1% $\pm$ 3.0%	+0.0pp	(near-linear)
Mean	75.4%	89.8%	+16.4pp	—

Reading the table. Six of eight domains improve. The two that do not (Breast Cancer, Wine) are near-linearly separable — the linear baseline is already at ceiling, and there is no nonlinear residual for the parsing layer to navigate. This is the expected behavior of a navigation layer: it does not consume the linear signal, it bridges between linear regions when such bridging is needed.

Fingerprint legibility. The retained families correspond directly to the known geometry of each domain. Circles activates radial and limit_cycle; XOR activates crossing and saddle; the cusp dataset activates the cusp primitive. The fingerprint is not arbitrary — it is a geometric diagnosis.

5. Real-World Tabular Data — A Boundary Condition

We applied the same protocol to five real-world tabular datasets: protein fold classification (38 sequences, SCOP 4-class), EEG eye-state (14-electrode neural signal, binary), steel annealing phase classification, ionosphere radar return, and DNA splice site detection.

Domain	LR(X)	LR(X+EBM)	Δ
Protein fold (SCOP)	48.3% $\pm$ 12.5%	48.3% $\pm$ 12.5%	+0.0pp
EEG eye-state	64.8% $\pm$ 1.7%	64.4% $\pm$ 1.6%	-0.4pp
Anneal phases	96.2% $\pm$ 1.1%	96.2% $\pm$ 1.1%	+0.0pp
Ionosphere radar	87.6% $\pm$ 3.2%	88.0% $\pm$ 3.0%	+0.4pp
DNA splice sites	82.3% $\pm$ 1.0%	82.3% $\pm$ 1.0%	+0.0pp

Near-zero gain across all five real-world domains. This is consistent with the central claim of Section 1: the Cherenkov EBM is not a competitor to linear regression on linearly separable data, and most real-world tabular datasets — after standard preprocessing — are nearly linearly separable.

The honest reading: the parsing layer activates appropriately on the few domains with diagnosable nonlinear structure (the Axolotl and saddle primitives appear on protein fold and EEG, where they would be expected), but does not translate to held-out accuracy gain at these sample sizes. The mechanism we diagnose is that the OLS weight fitter cannot reliably extract weak primitive signal from low-correlation residuals at 20-shot budgets. The remedy is not more primitives but better fitting — Section 6 introduces a

learned alternative.

6. V1 — Reinforcement Learning Over Primitives

We propose Verisimilitude V1: a tabular Q-learning protocol that treats the primitive selection problem as a Markov decision process. State is a discretized summary of the current residual field (variance, magnitude, skew, tail weight). Action is the choice of primitive family from the 13-element library. Reward is variance reduction at each step (dense shaping) plus held-out logistic regression lift at episode end (sparse terminal). Exploration follows epsilon-greedy with optimistic Q-initialization seeded by family priors.

The motivation is twofold. First, greedy synthesis cannot recover from a locally optimal primitive choice that blocks a globally better sequence. Q-learning amortizes this across episodes — the policy learns which primitive to deploy at which residual state. Second, terminal-reward credit assignment ensures that primitives are rewarded for downstream classification lift rather than for in-sample variance reduction alone, which directly addresses the real-world failure mode of Section 5.

Status of V1. Implementation is complete and verified end-to-end on synthetic and real-world data (510 lines, single file, scikit-learn compatible). Convergence behavior is consistent with theoretical expectation: the Q-table grows to roughly 50-80 unique states after 15-60 episodes, and the policy stabilizes around domain-appropriate primitives. The V1 pass criterion — (within-agreement & across-agreement & distinguishability & lift) — has not yet been satisfied on real-world data and remains the next concrete validation milestone.

7. The V1 Pass Criterion: Truth as Convergent Agreement

We define a working Verisimilitude 1 model as one that satisfies four criteria simultaneously on a real-world domain:

- (1) Within-agreement:** the fingerprint is stable across seeds on the same data. Multiple runs converge on the same primitive families.
- (2) Across-agreement:** the fingerprint is stable across related datasets (e.g., across subjects performing the same cognitive task).
- (3) Distinguishability:** the fingerprint differs meaningfully across different conditions or tasks.
- (4) Lift:** the augmented features improve held-out accuracy over the linear baseline.

This operationalizes a working definition of truth as *convergent agreement on togetherness in a linear context*. The fingerprint is the model's claim about the geometric structure of the domain. The four criteria test whether that claim is stable, transferable, discriminating, and useful. A model that passes all four is doing real structural inference — not pattern-matching to its priors and not overfitting to local geometry.

8. Position in the Literature

Versus existing EBMs. Standard energy-based models (Du and Mordatch, 2019; Grathwohl et al., 2020; Facebook EB-JEPA, 2024) parameterise the energy function with a neural network and train via contrastive divergence. Cherenkov replaces the neural backbone with a sum of named primitives, trading universal approximation for interpretability and a finite vocabulary. There is no Langevin sampling, no MCMC; synthesis is direct residual descent.

Versus JEPA. Joint Embedding Predictive Architectures (LeCun, 2022) predict in learned latent space rather than pixel space. Cherenkov shares this rejection of pixel-space prediction but predicts in *named* primitive space rather than learned latent space. The trade-off is expressiveness for legibility — the fingerprint can be read and audited.

Versus kernel methods. Radial basis function networks, kernel SVMs, and Gaussian processes also augment data with nonlinear basis functions. The Cherenkov primitive library is heterogeneous (13 functional forms, not 13 instances of one) and the selection is autonomous (no kernel hyperparameter tuning, no support vector selection). The system tells you *which* geometric structure it found, not just *that* it improved fit.

9. Limitations and Open Questions

Validation gap. All gains reported in Section 4 are on synthetic data with known nonlinear structure. The real-world results of Section 5 are consistent with the navigation-layer claim but do not yet demonstrate downstream utility on industrial data.

Sample efficiency. The OLS weight fitter is fragile at 20-shot budgets when primitive correlations are weak. V1 (Section 6) addresses this but is not yet validated.

High-dimensional scaling. The current implementation projects to 8D via PCA before primitive fitting. Native high-D synthesis is an open problem.

Vocabulary completeness. The 13-family library is a proposal, not a proof. Whether 13 primitives suffice to span the geometric structure of real-world domains is an empirical question.

10. Conclusion

The Cherenkov EBM is a geometric reasoning layer that operates between linear regression contexts. It does not compete with linear regression — it bridges between linear regimes via a finite alphabet of named geometric primitives. On synthetic geometric benchmarks where the bridging is required, the parsing layer recovers +16.4pp average accuracy gain across eight domains. On real-world tabular data where the bridging is not required, the layer produces near-zero gain — consistent with its intended role as a navigation layer rather than a universal nonlinear approximator.

Future work centers on V1 — reinforcement learning over the primitive library — and on the V1 pass criterion as a falsifiable test of whether the parsing layer produces stable, transferable, and discriminating geometric fingerprints on real-world data. The proposed test domain is neural signal: brain wave activation patterns across cognitive conditions, where the navigation-layer framing aligns directly with neural state transition geometry.

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*Code, package, and validation artifacts: [cherenkov.industries ·
github.com/psych1cparr0t-dev/cherenkov-ebm](https://cherenkov.industries.github.com/psych1cparr0t-dev/cherenkov-ebm)*